Programming with Python, Fall 2026

Course Project

The course project aims to explore an interesting problem of your choice using Python and a real-world dataset [1].

Form your own groups of **no more than 4 students** each, with a **maximum of 36 groups** in total. The grading criteria will not vary with group size, but each student's actual contribution will be taken into account.

This project requires three deliverables:

- Project report
- Code
- In-class oral presentation

See the "Report Template" section for the structure and required content of the project deliverables.

[1] Many public datasets are available online. You are encouraged to search for suitable datasets. For example, Kaggle Datasets: https://www.kaggle.com/datasets

1. Deadline

The course project report is due by **23:59 on November 13, 2026**. Please make sure to submit on time.

2. Requirements

- Each group may have **no more than 4 students**.
- The submission must include the report, complete code, and the experimental data used.
  - The report may be prepared in Word or LaTeX.
  - The course project must be submitted to both of the following:
    1. python123.io - the corresponding course unit

       Submit only the report and code to python123. **Do not submit data on this platform.**
    2. Email: python_projects@163.com
       - Each group needs to send **only one** project submission email.
       - Submit the report, code, and data used. Package them separately and upload them as attachments.
       - The code must run successfully. Experimental results must not be fabricated or altered.
       - Use the following email subject format:

         `CourseProjectFall2026-Name1-StudentID1-Name2-StudentID2-Name3-StudentID3-Name4-StudentID4`
       - In the email body, provide the project title and the names, student IDs, classes, responsibilities, and contributions of all group members.
- All groups shall deliver a **10-minute oral presentation** on their project during Weeks 9–11. All group members must be present on stage at the time of the presentation.

3. Project Content

Using Python and its ecosystem, complete an integrated, innovative project that applies artificial intelligence, data mining, or data analysis to your own field of study.

Requirements:

- Choose the functionality yourself and define a clear purpose for your program. Your code must contain **at least 100 lines** and use **at least two Python libraries** for artificial intelligence, data mining, or data analysis (e.g., scikit-learn, NumPy, pandas, Matplotlib, Seaborn, TensorFlow, and PyTorch). The project must include sufficient data analysis and data visualization results and demonstrate a degree of innovation.
- Write your project report using the template in Section IV.
- The code must contain sufficient comments to ensure readability.
- The code must run successfully, and the experimental results must be reproducible. Fabricating or tampering with experimental results is strictly prohibited.

4. Report Template

Include all sections below. Keep the writing concise and avoid repetition.

Report Title

Student Names, Student IDs, and Classes

1. Introduction

State the problem and its importance. Briefly summarize your approach and main findings.

2. Method

Justify your choice of models or algorithms in relation to state-of-the-art and commonly used methods. Describe your implementation and any improvements, using diagrams or algorithm outlines where helpful.

3. Experimental Results and Analysis

Describe the datasets and evaluation criteria. Present and interpret your results using clear tables and figures. Include additional analyses, such as hyperparameter robustness or ablation studies, where relevant.

4. Discussion and Conclusion

Summarize your findings and original contributions. Discuss strengths, limitations, lessons learned, and possible future improvements.

5. References

Cite all external sources, including theories, algorithms, models, data, and figures, using a consistent format.

Academic integrity: Do not present others' work as your own. Plagiarism will result in a zero grade for the course project.

Dataset Resources & Project Ideas

Choose a focused problem and a dataset that fits your computing resources, and skills. The examples below are starting points; develop your own question and analysis.

1. General Dataset Resources

UCI Machine Learning Repository - Tabular, text, and time-series datasets.

Kaggle Datasets - Community datasets across many topics; check source quality and access conditions.

Hugging Face Datasets - Language, vision, and other datasets; some require access approval.

World Bank Open Data and **Data.gov** - Development indicators and government open data.

Google Dataset Search and **Awesome Public Datasets** - Discovery resources; confirm access with the original provider.

2. Suggested Project Directions

Direction	Example dataset	Possible project
Tabular data	UCI Wine Quality	Predict wine quality; compare models and inspect errors.
Text analysis	UCI SMS Spam Collection	Compare spam classifiers and text representations.
Computer vision	Fashion-MNIST	Classify clothing images and analyze common mistakes.
Recommendation	MovieLens 100K	Compare popularity-based and collaborative recommenders.
Time series	UCI Bike Sharing	Forecast rental demand with chronological train/test splits.
Network analysis	SNAP Facebook	Explore communities and compare centrality measures.
Healthcare data	UCI Breast Cancer Wisconsin (Diagnostic)	Compare classifiers and feature selection for educational analysis.

3. Learning Resources

Use the official scikit-learn examples, pandas tutorials, and PyTorch tutorials for implementation guidance. Hugging Face Papers can help you find research ideas. Cite reused code and explain your own contributions.

4. Access and Project Scope

Check download access, data quality, licensing, and redistribution terms before selecting a dataset. Keep the project small enough to complete and reproduce.

For advanced projects, consider small subsets of ImageNet, COCO, or Open Images.